Research on Blockchain Secret Key Sharing and Its Digital Asset Applications

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Abstract

Public Key Infrastructure (PKI) is used in blockchain technology to authenticate entities and ensure the integrity of the blockchain. The overall security of a wallet depends on how anyone secures their private key, and the most critical data stored in a wallet is the user's private key, which is also the only identifier of ownership of encrypted digital assets. We often see cases of lost or stolen wallet private keys or even hacked cases in the domestic and international news, which makes people pay more attention to the issue of wallet private key management. Since there are some problems with traditional key management, such as cold and hot wallets and exchanges, it is no longer possible to fully guarantee the security of key management. All key management protocols have some weaknesses; if these weaknesses are not considered, the designed encryption tools are still insecure. For example, DH cannot protect against MITM (man-in-the-middle) and DoS (denial-of-service) attacks, and cryptosystems cannot be developed without proper protection against these attacks. To solve these challenges, we will propose research topics to improve the mechanism through smart contracts, which can achieve a more efficient approach. These research topics will result in a blockchain key management system with privacy, cross-trainability, and efficiency.

Keywords: Blockchain; Cryptocurrency; Key Management; Secret Sharing

1 Introduction

In recent years, with the rise of digital assets and virtual currencies, wallet security has received more and more attention. The private key is the only and most important line of defense for managing wallets. The private key may face the risk of being lost or even stolen. Once the wallet's private key is stolen or lost, we will likely not be able to get our digital assets back. Therefore, it is necessary to find a better solution to protect the security of private keys. Current tools for protecting private keys include Cold Wallet and Hot Wallet [11]. Although cold wallets can provide good protection, they are more restrictive in use and may also encounter problems with broken hardware, while hot wallets may be hacked—the risk of guest intrusion. In addition, some users are willing to host their assets in centralized institutions, such as cryptocurrency exchanges. However, this approach violates the decentralization and anonymity uniqueness of the blockchain [4, 8–10, 25, 29]. Among them, abnormal events in which assets are locked or lost prove that the security of exchange custody keys is questionable.

This research is based on the "Secret Sharing (SS)" private key partitioning proposed by Adi Shamir and George Blakley [16]. Assume that when users need to use private keys today, they can use the SS scheme to split the private keys. If it is just a general SS scheme, there will be a central person, which may lead to the problem of private key fragments being leaked or allowing the central person to crack it privately. If the SS scheme is improved through the smart contract of the blockchain, when the private key is needed, the requester can call this smart contract to generate the key and then distribute it. Or the user provides the master secret key, and then the system generates the subkeys and divides them. The subkeys can be restored according to the (t, n) setting to improve the above shortcomings. The following are the issues this research will explore and solve: secure storage after splitting private keys, seizure of illegal income, and inheritance and transfer of digital assets.

1) The problem of safe storage after private key division: The wallet's private key is essential. Once stolen or

lost, it will cause irreparable losses. Therefore, the security of the user's wallet private key must be ensured. This research topic must propose a practical solution by dividing private keys through smart contracts on the blockchain.

2) The problem of seizure of illegal proceeds from crypto assets: After supervisors seize the wallet keys of criminals, they may collaborate with other criminals to remove the keys and assets, or the supervisors may steal them themselves. This research will use the private key fragment decentralized encryption and decryption key hosting mechanism developed in the previous topic. Use blockchain smart contracts as a platform for private key custody to extend the scope of related applications. Therefore, this research topic will be applied to legalizing illegal gains from digital assets. After the inspectors seize the criminal's wallet key, it must be kept in custody. A more secure supervision mechanism can be achieved through smart contracts by dispersing keys among multiple people. This prevents criminal associates from removing keys or assets quickly and prevents supervisory personnel from guarding against theft.

3) The problem of inheritance and transfer of digital assets: If important information such as private keys is not provided to relatives when the holder passes away, they cannot transfer their encrypted assets to his family members. Therefore, it is hoped that through the smart contract mechanism, the key distribution can be set first and then triggered through the smart contract to achieve an effective mechanism for the custody of the key inheritance. The topic of this research is the application of digital asset inheritance, solving the problem of digital asset inheritance through smart contracts.

2 Main Contributions

This study has the following main contributions:

1) Solve the wallet private key theft problem that has occurred in recent years.

   In recent years, private key management has received much attention [3, 17, 21, 26]. It is a paramount personal privacy and confidential information. In addition, more and more people have begun to invest in cryptocurrency, NFT (Non-Fungible Token), etc., which has given rise to the problem of information security. For example, the wallet key is lost or stolen, mainly if used in a personal digital asset wallet. Therefore, a safer and more appropriate key management mechanism is needed. When storing management keys, it can provide higher security to ensure the security of private key management.

2) Improve the "Secret Sharing" private key split proposed by Shamir.

   Shamir proposed the "Secret Sharing" key sharing method (t, n). Private key splitting refers to dividing a private key (master key) into several fragments (as shown in Figure 1), and at least one part of them is required to restore the master key. This method can be used to prevent unauthorized persons from obtaining the entire private key to use it to sign or decrypt messages digitally [5–7,15,22,36,37]. Similar to the concept of jointly holding safe, multiple people need to hold and restore the private key before there is a way to unlock it and solve the problem of private key management. Since the administrator or a third party generates this private key division, this centralized operation will lead to the problem of being defrauded of the key. Therefore, smart contracts will be developed to improve the "Secret Sharing" private key division method proposed by Shamir. This research and development mechanism will be put on the chain to solve this problem.

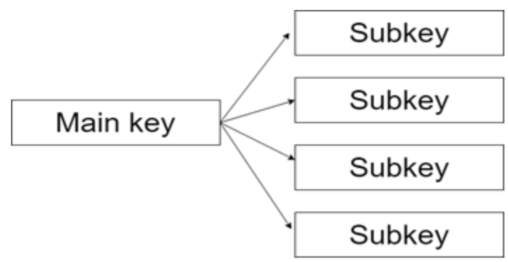

Figure 1: Properties of private key splitting

3) Solve the problem of private key custody through blockchain smart contracts.

   Most traditional key escrow has some flaws. As long as the device that stores the private key or the auxiliary tool for retrieving the private key is lost, the user can no longer retrieve the wallet. In addition, it is vulnerable to intrusion, allowing intruders to discover the key through any mechanism such as brute force cracking and weak encryption [27] or even the vicious collapse of the custodian or exchange [24], resulting in the inability to obtain the private key properly, of safekeeping. Therefore, this research is expected to use blockchain technology to prevent tampering and key leakage, protect the private key of the entire wallet, develop a better private key custody, achieve decentralization, and reduce the risk of being hacked.

4) Solve application scenarios related to the financial aspect of wallet assets.

   More and more criminal cases involving criminal crypto assets occur, which require the seizure of supervisory personnel. Due to the decentralization of cryptocurrencies, they cannot be directly seized like ordinary bank accounts. This type of legal seizure scenario of illegal gains (The criminal's private key has not been leaked out in a short time) uses public power to directly detain the criminal and transfer the assets to a new supervisory wallet. How to avoid supervisors colluding with criminals to transfer money or Transfers secretly are all urgent problems that digital asset finance needs to solve in the future. In addition, most of us in the Internet age have invested in or purchased cryptocurrency. Once the holder of these assets passes away, the assets cannot be transferred without leaving relevant information about the private key to relatives. This is a digital asset. Inheritance issues. This research will use the wallet control master key mechanism generated in the first year to develop automatic triggering smart contract technology to complete the transfer of assets.

3 Research Topics

Public key infrastructure (PKI) is used in blockchain technology to authenticate entities and ensure the integrity of the blockchain [27]. The overall security of a wallet depends on the security mechanisms protecting its private keys. The most important information stored in the wallet is the user's private key, also the unique identifier of the ownership of encrypted digital assets [23]. Cases of wallet private keys being lost, stolen, or even hacked often occur, making people pay more attention to the wallet private key management issue. Due to some problems in traditional key management, such as cold wallets, hot wallets, and exchanges, the security of key management cannot be fully guaranteed. All key management protocols have some weaknesses, and if these weaknesses are not considered, the designed encryption tool will still be insecure. For example, DH cannot protect against MITM (man-in-the-middle) and DoS (denial of service) attacks. Successful development of secure systems will not be possible without cryptographic systems that adequately protect against these attacks [1]. To solve these challenges, this study proposes a holistic solution: improve this mechanism through smart contracts to achieve a more efficient approach.

This study proposes three research themes:

1) First, use a blockchain smart contract to design the division and divergence of a key, and finally, implement the merger to ensure the security of key custody;

2) Under the framework of theme one, Extend related application scenarios in the blockchain part. The legal seizure of wallets illegally obtained from cryptocurrency allows the seizure of criminals' wallets to be appropriately supervised;

3) It is also a related application scenario, continuing the structure of theme one and improving it, triggering operations through smart contracts to solve the legacy of digital assets Referral question.

Integrating this research theme is a private, scalable, efficient blockchain key escrow system.

3.1 Research Topic 1: Research and Development of Blockchain Key Escrow

Most of the currently existing methods and traditional SS solutions require a central person to do the tasks of distributing and recovering keys. The SS scheme has a weakness in that dishonest participants cheat the share, and honest participants are forced to reconstruct fake secrets [33]. Therefore, it is necessary to split the private key fragments, store them with decentralized encryption and decryption, and perform key hosting. Pal *et al.* proposed a secure and efficient blockchain technology Group Key Management (GKM) framework [27]. In this GKM framework, a multi-layer architecture is assumed. The upper-level nodes have more privileges and rights than the lower-level nodes. At each level, there are multiple groups, each containing multiple nodes. Nodes belonging to the same group have the same permissions.

Private key segmentation is also used in many other areas, such as authentication and key management in the Internet of Things [28]. The number of devices connected to the Internet of Things is growing, and the data generated by these devices requires secure and effective access control mechanisms to ensure the privacy of users and data. Most traditional key management mechanisms rely on a trusted third party, such as a registry or key generation center, to generate and manage keys. Trusting a third party leads to a centralized architecture. Panda *et al.* solved these problems by designing a blockchain-based distributed IoT architecture that uses hash chains for secure key management. It is also used in intelligent transportation systems (ITS) and is an essential application of the Internet of Things. Encryption is an effective way to protect the confidentiality of data transmitted in ITS. Zhou *et al.* [38] proposed a method to implement threshold key management in blockchain-based ITS. The proposed threshold key management scheme is efficient and secure for multiple users in blockchain-based ITS, especially for material-sharing scenarios. Zhou *et al.* studied three secret sharing schemes: Shamir, Blakley, and CRT secret sharing schemes. It is concluded that the CRT-based scheme is more effective than the scheme based on Shamir secret sharing. In the system model, shared data is owned by the vehicle. Zhou *et al.* stored the data in the cloud for convenience of use and sharing. However, storing plaintext data may bring security and privacy issues, so these vehicles use private key segmentation to generate keys to encrypt the data, ensuring data storage security. A secret sharing scheme in the system generates keys, which are dispersed into parts and distributed to vehicles in a secure channel. In this way, only some vehicles can recover the key, each vehicle can control the data, and the data can be protected.

The main problem with systems based on blockchain technology is that users will not be able to access their keys due to device damage or loss [20]. Although blockchain technology is decentralized, key management has become the user's sole responsibility. This differs from the centralized approach, where the general account password has some administrator or IT person. The latter allows you to reset your password or create a new password if it is lost or forgotten. The current lack of this functionality in the system will result in the loss of assets. According to Chainanalysis, approximately 20% of Bitcoins (worth approximately $210 billion) have not been moved in the past five years and are therefore considered lost.

In the field of wallet and private key storage research, Boneh and Goldfeder *et al.* [2] proposed a threshold signature wallet scheme: the private key is divided into n fragments and stored by n participants. Completing each transaction requires more than the threshold t All participants jointly sign. Thota [34] proposed a software wallet that resides on the user's mobile device and uses the Hyperledger Fabric blockchain network. Jian [18] proposed a single-point failure protection scheme for blockchain wallets based on trustless central threshold elliptic curve digital signatures. This scheme can complete each transaction through the collaboration of multiple people, increasing the account itself to a certain extent: reliability and security. Dikshit [12] proposed a scheme based on participant identity, which provides different weights for participants.

Rezaeighaleh [32] proposed a new digital scheme for secure backup of hardware wallets that relies on display-enabled channel human visual verification on the hardware wallet. Gutoski [13] proposed a hierarchical deterministic (HD) wallet that effectively manages multiple private keys. Still, the hierarchical characteristics make the private keys have a fixed relationship and can have a certain degree of security and privacy. He *et al.* [14] proposed a new cryptocurrency wallet management scheme based on Decentralized Multi-Constrained Derangement (DMCD) to safely and stably store keys in a decentralized network. Wei *et al.* [35] proposed a new digital signature algorithm and used it to design an online wallet that can help users derive signatures without obtaining the user's private key.

With the rise of digital assets in recent years, wallet security has received much attention. As long as the private key is accidentally stolen or lost, severe losses will be caused. Therefore, custody can be achieved by dividing the private key, and effectively protecting the security and integrity of the private key is a research topic worth exploring. This research will solve the problems caused by general private key segmentation and propose feasible solutions to protect private keys. Designing a smart contract through the blockchain can avoid problems such as stealing a subkey or cheating to recover the key. This research topic needs to address the following questions:

1) Improve the private key segmentation of the SS scheme

   As far as the SS solution is concerned, today's private key hosting will have a central role. If executed through blockchain smart contracts, such problems can be avoided. This research can design a secure and private private key storage and hosting solution to solve this problem. The concept of the SS solution will be combined with the decentralized storage method of the blockchain to design a private key subkey storage solution that is both secure, private, and scalable.

2) Key management protocols have some weaknesses.

   The (t, n) threshold (Threshold) secret sharing scheme in the SS scheme can be used in different cryptographic applications. However, there will be some vulnerabilities, such as shared deception vulnerabilities and misbehavior by central actors. The main reason for this type of weakness is the lack of verifiability of the shares generated by the center. In addition, other key management protocols have some weaknesses, and the designed encryption tools are still insecure. For example, Diffie–Helman (DH) does not protect against DoS (Denial of Service) and MITM (Man-in-the-Middle) attacks. If these attacks are not adequately addressed, cryptographic systems will not be developed. In addition, the SS scheme has the weakness of dishonest participants cheating on their shares, and honest participants are forced to reconstruct fake secrets, etc. [23].

To summarize the above, this research project will solve two main problems:

1) Use blockchain smart contracts to design a framework for securely storing private key fragments so that both the user's private key and the private key fragments can be safely stored in Platform;

2) It can resist various attacks and improve existing vulnerabilities to ensure the security and integrity of users' private keys.

3.2 Research Topic 2: Research and Development to Prevent Illegal Income from Escaping Seizure of Virtual Currency

Using virtual currency as a criminal tool has become increasingly common. As an emerging encrypted virtual currency, the high risks of money laundering and crime hidden in virtual cryptocurrency have already attracted the attention of governments and judicial authorities in various countries. In the first half of 2022, tens of thousands of fraud cases have been reported, with financial losses exceeding 3 billion yuan. Fraudulent groups use cryptocurrency and other methods as money laundering tools, making it more difficult to recover lost amounts. In addition to new issues related to criminal law and criminal prosecution, asset recovery and seizure issues arise due to the asset value of cryptocurrencies. The main problem here is that cryptocurrencies are neither items nor rights. Confiscation need not be considered if the virtual currency units generated due to criminal conduct will likely lose value. Ensuring the security of Bitcoin during the investigation is also a matter of substantive significance.

Lee proposed using secret sharing [19]: split the gold key and then use the user's login password to generate the gold and trading platform keys after FIDO (Fast IDentity Online) authentication of the user's device. These keys are used as secret shared encryption keys after being unpacked, and at least two can be pushed back to the original keys to protect the risk of password loss or theft. After collecting relevant information on cryptocurrency trading platforms, this study found that the platform service provided by LoclEtherm is a more secure channel that allows users to control their keys, making it impossible for platform managers to obtain user keys. , all transactions are written on the blockchain, but it still fails to solve the problem of lost user keys. Therefore, it is proposed to use the FIDO standard authentication mechanism combined with the sharing method; it can also maintain security in addition to solving this problem.

The second research topic is based on the mechanism of the first research topic and develops the legal seizure of illegal income. Mainly because, in recent years, more and more crimes have been related to virtual currencies. Assume that the suspect's wallet is now seized, and the relevant police inspectors need to create a new wallet. The wallet will have a key, and you need to decide who to split the key to, then transfer the criminal's wallet to a new wallet for seizure. After the investigation, the money will be returned or compensated to the victim. This research topic will develop a legal seizure mechanism for illegal gains from virtual currencies to solve this problem. The narrative is as follows:

1) Solving the security issues of wallet custody

   After the wallet is seized, to prevent the criminal from conspiring with his accomplices, the private key is provided to the accomplices within a short period, and they take the opportunity to remove the illegal gains. Therefore, it is necessary to design adequate supervision to immediately transfer illegal gains to the supervisory personnel's supervision wallet and obtain reasonable control.

2) Solve the internal control problems of supervisors

   When a wallet is seized to prevent supervisors from stealing it, if only one person has the key to the wallet, the security of the seizure cannot be guaranteed. Therefore, a mechanism is designed to keep encrypted assets safe. Multiple people can hold private keys to solve the security problem of wallet seizure.

3.3 Research Topic 3: Research and Development on Secure Transfer of Digital Asset Heritage

Unlike in the real world, virtual crypto-assets do not disappear when their owner dies. The essential characteristic of encrypted assets is that only those who know the key can transfer assets. Actual ownership of a crypto asset is equivalent to a key, and conversely, everyone who knows the key is considered the legal owner of the relevant crypto asset. In general, a person cannot inherit their crypto assets. Because the keys disappear when the person dies, or the ownership of these crypto assets must be shared with a third party. However, the main problem is that legal owners can only transfer encrypted virtual assets. Conversely, nothing can be transferred as long as the holder dies, making the encrypted assets' inheritance complex [30, 31].

An essential characteristic of cryptoassets is that no one can transfer wealth except the owner of the corresponding key. To resolve inheritance issues, the easiest way is to write the private key into your will. However, this is not in the spirit of cryptocurrency because it relies on trusting a third party, which conflicts with the basic concept of cryptocurrency. Therefore, it is necessary to find ways to split private keys and strengthen and design smart contracts through the blockchain key custody mechanism developed by the research theme. Achieving the inheritance of crypto assets aligns with the original spirit of cryptocurrency.

This research topic will use the research topic 1 R&D mechanism to implement relevant applications. As we live in the digital age, after we pass away in the future, we hope to transfer digital assets to our loved ones through smart contract triggering, and we can make a will to decide in advance who we want to give it to. Who can use these? The triggering conditions for key distribution are activated through smart contracts. It will only be activated when the holder dies and is distributed to their respective wallets. To withdraw money later, these people need to present it simultaneously to achieve the digital asset distribution goal: fairness and integrity. However, as the data on the chain increases, the scale of the chain will become larger and larger, which will consume considerable costs and resources for the nodes.

4 Conclusion

This article has proposed three future research topics of blockchain secret key sharing and its digital asset applications. The first topic is the research and development of blockchain key custody. First, use blockchain to design an encrypted storage architecture to ensure patient privacy and integrity. Topic two is based on the structure of Topic one and extends to the blockchain part. Expand application field research on the legal seizure of illegal proceeds from virtual currency. The third topic is to solve the transfer of digital asset inheritance. Develop private key-splitting methods to make transferring digital assets on the blockchain more efficient. Therefore, integrating these three research yopics is a platform system that can be fully applied to user private key segmentation and can have privacy, security, and integrity.

This research will be able to achieve the following three goals:

1) Ensure the security of private key custody

   This project will research this issue and is expected to propose a solution to protect users' private keys and propose a framework that combines security, efficiency, and practicality.

2) Improve the problem of private key division and centralization

   The private key custody framework designed in this project is implemented through blockchain smart contracts to solve the problem of the central role of the traditional SS solution.

3) Solve the legal issues of seizure and custody of illegal gains from cryptocurrency and inheritance of digital assets

   This study predicts that the private key custody smart contract proposed in this study will be applied to the legal seizure and custody of illegally obtained cryptocurrency and the inheritance of digital assets, breaking through the scalability limitations and improving the applicability of the blockchain.

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